



Antony Meckhael

Software Engineer

Software Engineer with experience in backend development, distributed systems, data engineering, and AI-powered applications. Proven ability to build scalable services, end-to-end data pipelines, and intelligent systems. Co-author of published research in Industry 4.0 digital twins, with hands-on experience in microservices, Kubernetes, LLM-powered applications, real-time analytics, and mobile development.



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EDUCATION

Computer Engineering

GUC (German University in Cairo)

09/2021 - 06/2026

1.03 GPA (Excellent according to German scale)

Bachelor Thesis (A Graded)

- Titled "**Semantic-Driven Runtime Execution of Containerized Assets in Industry 4.0 Digital Twins**" —Developed at Universität Stuttgart, integrating BaSyx Java SDK with Spring Boot for conditional Docker container execution based on AAS metadata.
- Published as "[A Container-based Approach For Proactive Asset Administration Shell Digital Twins](#)" in collaboration with Universität Stuttgart professors, accepted and discussed at **EDTconf 2025**.

WORK EXPERIENCE

App Developer

Mentalee

05/2025 - 07/2025

Stuttgart, Germany

AI based Mental health platform

- Extended a Kotlin Multiplatform application with a focus on iOS integration, including architecture setup (Xcode, Swift bridges) and Unity embedding in the native iOS environment.
- Integrated TensorFlow models into the iOS app for on-device AI features, ensuring performance and reliability.
- Managed source control with GitLab and documented development results and code architecture.
- Collaborated in an agile feature team, contributing to the project's success through problem-solving, feedback, and team support.

FullStack Developer Intern

National Bank of Egypt

07/2024 - 08/2024

- Spearheaded the development and maintenance of a robust project using SQL Server Management Studio (SSMS), ASP.NET, Entity Framework, and Xamarin.Forms, reducing system downtime by 20%.
- Partnered with cross-functional teams to deliver scalable solutions, ensuring a 95% client satisfaction rate.
- Conducted comprehensive debugging and optimization, enhancing application performance by 30%.
- Gained valuable insights from experienced mentors, strengthening both technical and communication skills.

SKILLS

- **Programming Languages:** Java, Python, Kotlin, Swift, C#, JavaScript, SQL.
- **Backend & Distributed Systems:** Spring Boot, REST APIs, Microservices, RabbitMQ, PostgreSQL, MongoDB, Redis, Elasticsearch, Neo4j, Cassandra.
- **Data Engineering:** PySpark, Kafka, Airflow, PostgreSQL, MongoDB, ETL Pipelines.
- **AI & Machine Learning:** Scikit-learn, TensorFlow, LightGBM, NLP, LLMs, RAG, Graph-RAG, SHAP, LIME
- **Cloud & DevOps:** Docker, Kubernetes, GitHub Actions, Grafana, Prometheus, Loki, Git, Maven

PERSONAL PROJECTS

Scrum Master for Food Delivery Platform

- Led a team of 14 engineers in evolving a Spring Boot monolith into a 5-service Kubernetes-based microservices platform.
- Developed core backend components including a reactive JWT API Gateway, shared contracts module, and payment services.
- Implemented RabbitMQ choreography sagas and polyglot persistence using PostgreSQL, Redis, MongoDB, Elasticsearch, Neo4j, and Cassandra.
- Built production-style infrastructure with Docker, Kubernetes, GitHub Actions CI/CD, Prometheus, Grafana, and Loki.

Real-Time Stock Analytics Platform (Data Engineering)

- Designed and implemented a real-time data pipeline using Apache Kafka and Spark (PySpark) to process high-velocity trading data.
- Orchestrated distributed ETL workflows with Apache Airflow and deployed services using Docker Compose.
- Built a PostgreSQL data warehouse and a Streamlit-based natural language-to-SQL analytics interface.
- Ensured reproducible, scalable execution through containerized architecture.

Intelligent Hotel Analytics & Graph-RAG System

- Built an end-to-end machine learning pipeline on real hotel, user, and review data, including cleaning, feature engineering, class imbalance handling (SMOTE), and reproducible inference artifacts.
- Detected and resolved severe target leakage using permutation importance and SHAP, validating model behavior with global and local explainability (SHAP, LIME).
- Trained and compared Logistic Regression, Random Forest, LightGBM, and FFNNs, tuning hyperparameters to improve generalization and macro-F1 performance.
- Designed a Neo4j Knowledge Graph and integrated hybrid retrieval (Cypher + embeddings) to ground LLM responses in structured data (Graph-RAG).

Scrum master for Comprehensive Tourism Management Platform (Ranked 1st in Course Project Evaluation)

- Led a team of 10 as Scrum Master, driving Agile practices to ensure on-time delivery of the platform built using the MERN stack (MongoDB, Express.js, React, Node.js).
- Developed a multi-functional web application allowing tourists to book flights, hotels, activities, and tickets, while enabling sellers, tour guides, and advertisers to access tailored functionalities.
- Designed an admin dashboard for efficient content management, user access control, and advertisement management, ensuring smooth operations.

LANGUAGES

- **English:** Fluent
- **Arabic:** Native
- **German:** Intermediate